Galois Theory: Intuition Behind Field Extensions

— *EYNTKA* Series —

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This blog is an extract from EYNTKA Algebra [3] going over my intuition on why field extensions are the language of Galois theory.

For eons, mathematicians looked for relations between coefficients and roots of a polynomial. The answer is “trivial” for linear polynomials, given:

$$ax + b = 0 \qquad \text{with solution } \alpha \text{, then} \quad \alpha = -ba^{-1}$$

For a quadratic $x^2 + bx + c$ with roots α and β, then $\alpha + \beta = -b$ and $\alpha\beta = c$[1], and each individual root is given by the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

More generally, a lot of focus was put into finding the roots of polynomials given their coefficients[2], eventually finding the cubic and quartic equation relating the coefficients to roots of cubic and quartic polynomials respectively. However, the effort to generalize to higher degree polynomials brought limited results, in particular even for the next degree (the quintic) it was unclear how its roots can written in terms of it’s coefficients and the arithmetic operations of $+, -, \times, \div$ and $\sqrt{-}$. It wasn’t until the turn of the 18th century when serious progress has been made on this front. The key insight was to treat the coefficients of a polynomials as determining a dynamical system (that is, the dynamic of how the roots of a polynomial change as we move around the coefficients). In [2, the chapter on field extensions], the right context to consider is the algebraic closure of a field; for pedagogical purposes we stick to $\overline{\mathbb{Q}}$ or $\mathbb{C}$ namely since this is the theories starting point and generalizations to $\mathbb{F}_{q^n}$ and other fields/rings is the natural progression.

[1] See [2, the section on symmetric functions] for much more detail about these relations

[2] This is a little anachronistic, for when the quadratic formula was first discovered in Babylonia and more succinctly by Brahmagupta, the “modern” notion of polynomial and variables where not used, but rather it was shown by relating areas of circles and rectangles

To demonstrate these dynamics, consider $x^2 + x - 1 = 0$. This equation (by using the quadratic equations) has solutions:

$$x_1 = \frac{-1+\sqrt{5}}{2} x_2 = \frac{-1-\sqrt{5}}{2}$$

If the b and c coefficients are swapped to get $x^2 - x + 1 = 0$, then the solutions are:

$$x_1 = \frac{1+i\sqrt{3}}{2} x_2 = \frac{1-i\sqrt{3}}{2}$$

In other words if (a, b, c) represents the coefficients, then the permutation action $(2\ 3) \in S_3$ does not leave the roots invariant, that is this is *not* invariant to coefficient permutation. Note these dynamics are "discrete", but they are the shadow of a continuous one. Let $\mathcal{P}_n$ denote the space of monic degree-n complex polynomials with distinct roots, that is, $\mathbb{C}^n$ with the discriminant locus removed. Sending a polynomial to its unordered set of roots exhibits the space of *ordered* root tuples as an n-sheeted covering of $\mathcal{P}_n$, so a *loop* of coefficients returns the root set to itself while permuting the individual roots. This is a homomorphism

$$\pi_1(\mathcal{P}_n) \longrightarrow S_n$$

and the permutation above is one of its values: the discrete action is what a continuous loop leaves behind. The group $\pi_1(\mathcal{P}_n)$ is the braid group B_n, and for a generic polynomial the image of this map is all of S_n, the same S_n that will shortly appear as a Galois group. That a fundamental group turns up here is not an accident; see box 0.0.1.

Now, given a general $ax^2 + bx + c = 0$, factoring as $(x - \alpha)(x - \beta)$, then by commutativity:

$$(x - \alpha)(x - \beta) = (x - \beta)(x - \alpha)$$

In other words, applying an S_2 action on the roots leaves the coefficients invariant[3 4].

The first immediate introduction of dynamics into polynomials is by applying group-actions to the roots. The realization that links roots to coefficients is that, given we can completely split a polynomial:

$$x^n + \cdots + \alpha_1 x + \alpha_0 = (x - a_1)(x - a_2) \cdots (x - a_n)$$

each coefficient of the original polynomial can be represented as a *symmetric function* (see [2, the section on symmetric functions]).

$$\begin{aligned}
\alpha_{n-1} &= -(a_1 + a_2 + \cdots + a_n) \\
\alpha_{n-2} &= a_1 a_2 + a_1 a_3 + \cdots + a_1 a_n + a_2 a_3 + \cdots a_{n-1} a_n \\
&\cdots \\
\alpha_0 &= (-1)^n a_1 a_2 \cdots a_n
\end{aligned}$$

the signs alternating, exactly as in the quadratic case above where $\alpha + \beta = -b$ and $\alpha\beta = c$. These form a natural relationship between the roots and the coefficients. It was noticed that these equations are invariant under permutation action of S_n; if $A = \{A_0, ..., A_{n-1}\}$ represent the above n equations

[3] Concretely, take $x^2 - c$, whose roots are $\pm\sqrt{c}$. As c travels once around the circle $c = re^{i\theta}$, with θ running from 0 to 2π, the roots $\pm\sqrt{r}e^{i\theta/2}$ trade places: a single loop of the coefficient around the discriminant realises the transposition in S_2

[4] As a consequence, there is no continuous function that can take in coefficients and return roots

and the action is given by $(r\ s) \cdot a_r = a_s$, then $\sigma \cdot A_i = A_i$ for each A_i and σ, hence the action $S_n \curvearrowright A$ is the trivial action. Thus, there is always a natural group-action.

As any roots of polynomial induce such equations and induce this trivial action, these don't give enough information about polynomials. It shall be shown that roots of a generic polynomial satisfy no further equation, and in these cases there is nothing more that can be done to understand a polynomial via a group-action on its roots. However, there are polynomials whose roots shall have *more* relations the roots satisfy, limiting what group can act on them (i.e. a subgroup of S_n will act trivially, but S_n will not). This chapter shall make rigorous this realization and translate it into the modern language of field theory.

This realization almost managed to show the general quintic is unsolvable:

> "The quintic was almost proven to have no general solutions by radicals by Paolo Ruffini in 1799, whose key insight was to use permutation groups, not just a single permutation. His solution contained a gap, which Cauchy considered minor, though this was not patched until the work of the Norwegian mathematician Niels Henrik Abel, who published a proof in 1824, thus establishing the Abel-Ruffini theorem."[5]

This theory was established with the aim to show there is an unsolvable quintic, but not how to find whether a given quintic is solvable. This is where the work of Évariste Galois comes in. He noticed that the key insight was to work with all possible relationships of the roots of the polynomial, assuming all the roots exist.

Modern uses of Galois Theory

Galois groups are central in both algebraic geometry and number theory, where they serve different purposes in each domain. The galois group of a polynomial:

1. establishes the algebraic relations between the roots and creates a "covering space" that has some similar properties to the covering spaces seen in algebraic topology (see remark 0.0.1),
2. establishes the invariance necessary to find a solution over a lower field, in particular if L/K is a field extension and a root of an $L[x]$ polynomial has a root in K, then the root is invariant under the $\mathrm{Gal}_K(L)$-action.

From action on roots to field theory

Take the polynomial:

$$x^4 - 2 = 0$$

which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion ([2, Eisenstein's criterion]). Adjoining $\sqrt[4]{2}$, $\mathbb{Q}(\sqrt[4]{2})$, the polynomial splits to:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x^2 + \sqrt{2})$$

Adjoining $\sqrt[4]{2}i$, the polynomial splits completely:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - \sqrt[4]{2}i)(x + \sqrt[4]{2}i) \tag{1}$$

[5]taken from Wikipedia for Galois Theory

Already, there is some interesting phenomena at play, for example, another way of getting this splitting is by adjoining $\sqrt[4]{2}$ and then adjoining i (or adjoining i then $\sqrt[4]{2}$). In fact, by just adjoining $\sqrt[4]{2}+i$, we immediately get this splitting. Indeed, certainly $\mathbb{Q}(\sqrt[4]{2}+i) \subseteq \mathbb{Q}(\sqrt[4]{2}, i)$. Conversely, to show $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, letting $a = \sqrt[4]{2}+i$, then:

$$\begin{aligned} a - i &= \sqrt[4]{2} \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ (a-i)^4 &= 2 \\ \Leftrightarrow\ a^4 - 4a^3 i - 6a^2 + 4ai + 1 &= 2 \\ \Leftrightarrow\ i(4a - 4a^3) &= 1 - a^4 + 6a^2 \\ \Leftrightarrow\ i &= \frac{1 - a^4 + 6a^2}{4a - 4a^3} \end{aligned}$$

showing that $i \in \mathbb{Q}(\sqrt[4]{2}+i)$, and so $a - i = \sqrt[4]{2} \in \mathbb{Q}(\sqrt[4]{2}+i)$. More generally for any finite separable extension $K/\mathbb{Q}$, such an element always exists and is constructible by the primitive element theorem ([2, the primitive element theorem]).

Let us understand the equations the roots form given by symmetric functions. As $x^4 - 2 = x^4 + 0x^3 + 0x^2 + 0x - 2$, the only meaningful relation obtained by multiplying out the roots in (1) is:

$$(\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) = -2 \qquad \text{equiv.} \qquad (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) + 2 = 0$$

This equation is invariant under the action of S_4, namely if

$$(a_1, a_2, a_3, a_4) = (\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}, -\sqrt[4]{2}i)$$

represent the 4 roots, and $A_0 = (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) + 2 = 0$, then:

$$\sigma \cdot A_0 = A_0 \qquad \forall \sigma \in S_4$$

However, there are more equations that can be formed with these roots (using the operations of a field). With the labelling above, the first and third roots are negatives of one another, as are the second and fourth:

$$\begin{aligned} A_1 :\ a_1 + a_3 &= \sqrt[4]{2} + (-\sqrt[4]{2}) = 0 \\ A_2 :\ a_2 + a_4 &= \sqrt[4]{2}i + (-\sqrt[4]{2}i) = 0 \end{aligned}$$

These equations are *not* invariant under the permutation action of S_4:

$$(1\ 2) \cdot A_1 \neq A_1 \qquad (\text{concretely: } (1\ 2) \cdot (a_1 + a_3) = a_2 + a_3 = \sqrt[4]{2}i - \sqrt[4]{2} \neq 0)$$

A subgroup of S_4 must be chosen, namely notice that $D_4 \cong \langle (1\ 2\ 3\ 4), (1\ 3) \rangle \subseteq S_4$[6] keeps them invariant. It does not fix each one separately: the four-cycle carries A_1 to A_2 and back. What it preserves is the *pairing* $\{\{1,3\},\{2,4\}\}$ of the roots into negative pairs, and so the pair of equations as a set:

$$D_4 \cdot A_0 = A_0 \qquad D_4 \cdot \{A_1, A_2\} = \{A_1, A_2\}$$

[6] The reader may recall [2, the exercise on group presentations] asks to show S_n is generated by a two cycle and an n-cycle; it was important the 2-cycle was an *adjacent* cycle

Indeed D_4 is precisely the stabiliser of that pairing inside S_4, and it has order $8 = [\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}]$, the first hint that this is the right subgroup. Are there any other equations that we can be considered which would further restrict the choice of invariant permutation action? To answer this, the above is re-framed in the language of field extensions.

Let us say f is a polynomial over $\mathbb{Q}$ and that α is a root of f. Then the algebraic extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains *all* possible polynomial equations that can be generated with elements from $\mathbb{Q}$ and α (recall $\mathbb{Q}(\alpha) = \mathbb{Q}[\alpha]$, see [2, the description of elements of a finite simple extension]). If any of these polynomials happen to be another root of f, that is $p(\alpha) = \beta$ where $f(\beta) = 0$ and $p(x) \in \mathbb{Q}[x]$, then $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains this relationship between the roots. Let's say one of these relationships is represented as :

$$P(x) = p(x) - \beta \text{ so that } P(\alpha) = p(\alpha) - \beta = 0$$

Then a $\mathbb{Q}$-automorphism (in this case nothing more than a field automorphism) would have to preserve this relationship (it must map roots to roots). Hence, the set of $\mathbb{Q}$-automorphism captures the relationships between α and all possible roots that are $\mathbb{Q}$-polynomials of α. If $K/\mathbb{Q}$ is the splitting field of f, then the $\mathbb{Q}$-automorphisms of K would be restricted to automorphism that must satisfy all possible relationships between roots. Given $K/\mathbb{Q}$ is a splitting field for f, as any $\mathbb{Q}$-automorphisms σ is determined by where the roots map to, it is associated to a permutation in S_n. If there were no relationships (you need to add n roots to split f), then the automorphism group would be isomorphic to S_n. If there is any other type of group, that means there are some relationships between the roots (just like the example of $\mathbb{Q}(\sqrt[4]{2} + i)/\mathbb{Q}$).

The automorphism group of the splitting field above is called the *galois group* and it is the aim of this chapter define the galois group for any field extension L/K, to be able to find some galois groups, to find how frequent certain galois groups are, to show the correspondence between field extension and galois groups, and answer some long-pending mathematical questions such as the solvability of [general] quintic using the newly developed theory.

Another way of thinking about galois group is they are a rough classification tool for field extensions: if two field extensions have different Galois groups they cannot be the same extensions. It is similar in roughness to the fundamental group $\pi_1(X)$ as is often encountered in algebraic topology. To see this roughness in classification, as the splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$ of $x^4 - 2$ is equal to $\mathbb{Q}(\sqrt[4]{2} + i)$, and the minimal polynomial of $\sqrt[4]{2} + i$ is:

$$x^8 + 4x^6 + 2x^4 + 28x^2 + 1$$

then they share the same restrictions on their roots as they have the same galois group, but they certainly also have many differences (for example, these two polynomials have different discriminants, see [2, the definition of the discriminant]).

0.0.1 Galois Group as Fundamental Group

(Assuming strong familiarity with algebraic geometry and algebraic topology) Under the right framework in algebraic geometry, galois groups can be interpreted as *étale fundamental group* for $\operatorname{Spec} K$,

$$\pi_1^{\text{ét}}(\operatorname{Spec}(K)) \cong \operatorname{Gal}_K(\overline{K})$$

The name *fundamental group* is well-deserved for if X is a connected $\mathbb{C}$-scheme of finite type (i.e. "essentially" a connected variety) and $X(\mathbb{C})$ is its analytification, then:

$$\pi_1^{\text{ét}}(X) \cong \pi_1^{\text{top}}\widehat{(X(\mathbb{C}}))$$

where $\pi_1^{\text{top}}(-)$ is the usual fundamental group and $\widehat{(-)}$ is the profinite completion with respect to its finite-index normal subgroups. See [1] for the details.

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